Estimating Maximum Usable Frequency of HF Communication Links Using the Low-Latitude Long-Range Ionospheric Radar

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139
Full
Text Views

Abstract

Document Sections

I. Introduction

II. Instruments and Observations

III. MUF Estimation for Given Link

IV. MUF Map Estimates for Fan Sectors

V. Discussion

Show Full Outline

Authors

Figures

References

Keywords

Metrics

More Like This

Abstract:
High-frequency (HF) radio communication is essential for long-range connectivity in remote and maritime regions lacking conventional communication infrastructure. The maximum usable frequency (MUF), which is critical to HF effectiveness, is sensitive to dynamic ionospheric conditions. Traditional MUF observations and global models suffer from limited coverage or reduced accuracy, particularly over low-latitude oceanic regions where ionospheric variability is pronounced. This study introduces a practical MUF inversion method for low-latitude regions, utilizing ground/sea scatter echoes from the low-latitude long-range ionospheric radar (LARID) at Dongfang (19.2°N, 108.8°E). By combining ray-path geometry with ionospheric characteristic parameters, the proposed technique generates MUF profiles through nonlinear fitting of ground distance and operating frequency. This method demonstrates better performance in low-latitude regions compared to the linear fitting approach traditionally used by HF radars at high latitudes. Validation against ionosonde data at Ledong (18.4°N, 109°E) shows close agreement in diurnal variation trends. The results reveal MUF asymmetries driven by the equatorial ionization anomaly (EIA) and local time effects—features that cannot be captured by empirical models or single ionosonde measurements. This approach addresses a critical gap in real-time MUF monitoring over oceanic regions, enhancing the reliability of HF communication links in maritime and emergency scenarios.

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